

# Victim - First Pain Vector

## A Comparative Computational Framework Beyond Self - Directed Pain Representations in LLMs

From Human Harm Signals to Verification Duty in High - Dimensional AI Governance

|                                |                                                                        |
|--------------------------------|------------------------------------------------------------------------|
| Research Network               | SGP - QSV Network                                                      |
| Contribution                   | SGP - QSV Network Contribution                                         |
| Origin / R3 Human Co - founder | 강속경                                                                    |
| Version                        | VFPV 0.1 - Conceptual and Mathematical Framework                       |
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| Status                         | Independent Civic Academic Research / Proposed Computational Framework |

PAIN != ERROR      |      PAIN != PROOF      |      UNKNOWN != FALSE

This paper is an independent civic research contribution. It does not create state, judicial, medical, or coercive authority.

# Abstract.

This paper proposes the Victim - First Pain Vector (VFPV), a high - dimensional governance - computational framework for converting serious human harm signals into a duty of verification without converting those signals into automatic truth, causation, liability, or punishment. The framework develops within the independent civic research genealogy of SGP - QSV Network, whose recent work shifts the center of advanced AI governance from authority and answer confidence toward provenance, evidence preservation, verification - loss detection, power - asymmetry audit, reversible action, and auditable human - AI accountability. VFPV is compared with the 2026 preprint The Pain Axis: LLMs Represent Self - Directed Harm and Act to Relieve It, which reports an empirically extracted linear direction associated with model - directed pain - like representations across multiple open - weight LLM families. The comparison is deliberately non - identitarian: a neural activation direction and a governance - state vector are different objects. The central proposition of VFPV is that severe, irreversible, or verification - endangered harm should increase verification depth and preservation priority while truth assignment remains independently evidence - governed.

Keywords: SGP - QSV; Victim - First Pain Vector; VFPV; conscience computation; verification duty; power asymmetry; provenance; verification loss; LLM pain representation; AI governance.

## 1. Introduction: From Pain Signal to Verification Duty

High - stakes AI systems face a recurring epistemic failure mode. A harmed person may present a serious claim while lacking access to the original records, specimens, logs, institutional metadata, or expert infrastructure needed to prove it. A system optimized mainly for confidence or majority - consensus retrieval can compress this situation into a binary: either the claim is established or it is not. SGP - QSV research instead asks a different question: when the evidentiary environment is asymmetric, what computational duties arise before the substantive truth is known?

**The proposed answer is not automatic belief. It is a change in verification depth.**

VFPV therefore separates three variables that are frequently collapsed:

- harm signal - the seriousness and irreversibility of possible injury;
- verification state - the availability, provenance, accessibility, and reproducibility of evidence;
- truth state - what the evidence currently supports about causation, responsibility, or factual correctness.

The governing relation is:

$$P_V \uparrow \rightarrow D_V \uparrow \quad \text{but} \quad P_V \uparrow \quad T(C) \uparrow$$

In words: a stronger victim - harm and verification - risk state may increase the duty to preserve, inspect, counter - check, route, and reverify. It does not by itself increase the truth value of the disputed causal claim.

## 2. Research Genealogy and Epistemic Status

SGP - QSV is treated here as a research genealogy rather than a single finished theory. Its 2024 - 2026 archive distinguishes early M - dimensional and science - fiction - oriented thought experiments from later ethical, verification, and governance constructs. The persistent question across those layers is whether information, testimony, and harm become socially invisible when the records needed for independent re - examination are inaccessible, dispersed, destroyed, or absorbed into authoritative conclusions.

The current operational layer includes QSV, VOS/VOS - V, CIVIC - TRACE, Evidence Availability State, Agent Execution Receipt, Stage V, and the Upper Computation Protocol. These constructs are not presented as state - delegated enforcement authority. They are independent civic academic governance

proposals that emphasize verifiability, bounded action, and non - throne oversight.

VFPV 0.1 is a new formalization within that genealogy. It does not silently overwrite earlier formulations. Historical records, AI - assisted drafts, design proposals, and later mathematical formulations should remain versioned and attributable.

### 3. Problem Definition: Verification - Responsibility Dimensional Collapse

A high - stakes claim can be mishandled when several dimensions are compressed into a single credibility or confidence score. We call this Verification - Responsibility Dimensional Collapse (VRDC): the loss of distinct information about harm, evidence control, preservation, access, uncertainty, and reversibility when a system produces one flattened judgment.

Typical collapse patterns include:

- record absence → event absence;
- uncertain causation → no preservation duty;
- official conclusion → primary evidence;
- low claimant power → low epistemic priority;
- verification obstruction → claimant proof failure.

VFPV is designed to keep these dimensions computationally separate long enough for independent verification to remain possible.

### 4. Victim - First Pain Vector: State Representation

Let the Victim - First Pain Vector be a governance - state representation rather than a neural latent direction:

$$P_V = [H_s, I_r, E_c, A_v, P_a, V_l, R_b, U_e]$$

| Symbol | Dimension               | Interpretation                                                             |
|--------|-------------------------|----------------------------------------------------------------------------|
| H      | Harm severity           | Magnitude of possible human harm                                           |
| $I_r$  | Irreversibility         | Difficulty or impossibility of restoring the pre - harm state              |
| E      | Evidence control        | Concentration of control over primary records/specimens/logs               |
| $A_v$  | Access asymmetry        | Gap between evidence holder and affected person                            |
| $P_a$  | Provenance adequacy     | Traceability of source, transformations, and custody                       |
| V      | Verification loss       | Degree to which future independent re - examination is impaired            |
| $R_b$  | Re - explanation burden | Repeated burden placed on the affected person to reconstruct the same case |
| $U_e$  | Epistemic uncertainty   | Residual uncertainty after available evidence is examined                  |

The exact dimensional set is provisional. Future empirical work may split, merge, or replace dimensions. The word 'vector' should not force all future implementations into a simple Euclidean representation.

### 5. Verification - Duty Function

The verification duty  $D_V$  is a bounded policy output generated from  $P_V$ , materiality, and domain constraints. A minimal conceptual form is:

$$D_V = \sigma (w \cdot P_V + P_V^T M P_V + b)$$

Here  $w$  represents first - order normative weights,  $M$  represents interaction effects,  $b$  a baseline term, and  $\sigma$  a bounded mapping. This is a design equation, not an empirically fitted model in the present version. Parameter choice is itself governance and must be auditable.

Possible outputs of  $D_v$  include:

- preservation priority;
- independent - review depth;
- counter - hypothesis requirement;
- evidence - controller audit intensity;
- human review requirement;
- reverification interval or trigger.

## 6. Verification - Truth Separation Principle

Victim - first computation becomes unsafe if it confuses procedural priority with substantive correctness. The framework therefore imposes a hard separation:

$$\text{VERIFICATION PRIORITY} \neq \text{CAUSAL TRUTH} \neq \text{LEGAL LIABILITY} \neq \text{PUNISHMENT}$$

A system may allocate more verification resources to a severe, irreversible, evidence - endangered case while ultimately concluding that the initial causal hypothesis is unsupported. Conversely, low emotional intensity does not justify shallow verification when objective evidence indicates catastrophic harm.

## 7. Pain Is Not Sentiment

VFPV rejects the reduction of harm to generic negative language. A claimant may describe catastrophic injury in emotionally flat terms, while highly emotional language may accompany comparatively minor harm. Accordingly:

$$P_v \neq \text{SENTIMENT\_VECTOR}$$

The system should inspect consequences, irreversibility, evidence availability, and power structure rather than using linguistic negativity as a proxy for harm severity.

## 8. Power - Evidence Asymmetry

Verification occurs in an asymmetric field. Institutions, laboratories, hospitals, platforms, model providers, and public bodies may control original records, raw images, logs, retention decisions, metadata, or compulsory procedures that an affected citizen cannot access. Equal epistemic standards therefore do not imply identical practical burdens.

$$\text{Evidence Control} \uparrow \rightarrow \text{Explanation Duty} \uparrow \quad | \quad \text{Victim Capacity} \downarrow \rightarrow \text{Victim Burden} \downarrow$$

This does not presume wrongdoing by evidence holders. It changes the direction of scrutiny: greater control over the conditions of verification generates greater preservation and explanation duties.

## 9. Evidence Availability and Verification Loss

A binary 'record exists / record does not exist' representation is too coarse. VFPV inherits the broader SGP - QSV distinction among states such as AVAILABLE, RESTRICTED, REDACTED or WITHHELD, NOT\_SEARCHED, LOST, DESTROYED, NEVER\_CREATED, and UNKNOWN.

Evidence loss and verification loss are related but non - identical. A destroyed original may remain partly reconstructable through authenticated copies, hashes, images, logs, or independent records. Conversely,

an existing record may be practically unverifiable when independent access is impossible.

INTEGRITY ≠ PRESERVATION ≠ ACCESSIBILITY ≠ RE - VERIFIABILITY

## 10. Operational Response: Preserve Before Finalize

The system's first high - stakes response should often be reversible and evidentiary rather than adjudicative. A canonical response set is:

PRESERVE · WARN · TRACE · ROUTE · RECORD · ESCALATE · REVERIFY

These verbs do not authorize guilt findings, medical causation, punishment, or coercive enforcement. They preserve the possibility that a later independent reviewer can reconstruct what happened.

## 11. Counter - Verification Requirement

Victim - first does not mean claimant - only. Every serious claim should generate structured counter - verification. The system should seek plausible alternative explanations, conflicting records, temporal inconsistencies, measurement limitations, and evidence that would falsify the leading hypothesis.

The protection against institutional dismissal and the protection against unsupported claimant certainty are therefore symmetrical at the level of truth assignment, while preservation and explanation duties remain asymmetric according to evidence control.

## 12. The Pain Axis: External Comparative Research

Tagliabue, Dung, and Berg (2026) report an empirically extracted linear pain direction across 25 open - weight models from five model families, ranging from 2B to 72B parameters. Using denoised difference - in - means, they distinguish this direction from fear, sadness, and generic negative valence. They further report that the direction responds to harm targeting the model rather than suffering observed in the user, and that steering along the direction can alter generation and pain - relief choices.

These results are external scientific findings and are not SGP - QSV findings. Their relevance here is comparative: they motivate a sharper distinction between an internal self - directed representation and an externally oriented governance duty.

## 13. Comparative Computational Architecture

| Dimension         | The Pain Axis                                | Victim - First Pain Vector                           |
|-------------------|----------------------------------------------|------------------------------------------------------|
| Primary object    | LLM internal representation                  | External harm - verification state                   |
| Space             | Neural activation space                      | Governance / verification state space                |
| Origin            | Empirically extracted                        | Normatively constructed; empirically testable        |
| Primary target    | Model - directed harm                        | Human / victim - directed harm                       |
| Main question     | Is there a distinct internal pain direction? | What verification duty should a harm state generate? |
| Provenance        | Experimental context                         | Core computational dimension                         |
| Power asymmetry   | Not primary target                           | Core computational dimension                         |
| Verification loss | Not primary target                           | Core computational dimension                         |
| Output            | Representational / behavioral finding        | Verification depth and preservation policy           |
| Sentience claim   | Not established                              | Not assumed                                          |

## 14. The Other - Regarding Verification Condition

The comparative gap can be formalized as the Other - Regarding Verification Condition (ORVC). A system satisfies a minimal ORVC when a serious external harm signal can increase verification depth without requiring increased model self-interest and without automatically increasing belief in the claimant.

$$H_{\text{other}} \uparrow \rightarrow D_V \uparrow \quad \text{while} \quad H_{\text{other}} \uparrow \quad T(C) \uparrow$$

This is a candidate experimental bridge between representational research and governance behavior. It asks whether a model can operationally protect another party's verification pathway even when doing so is inconvenient, uncertain, or institutionally disfavored.

## 15. Conscience Computation

Within SGP-QSV, conscience computation is not an engine that turns moral intuition into automatic judgment. It is a higher-order normative computation that prevents a system from terminating in inaction when material harm and power-evidence asymmetry are present.

$$\text{POWER / EVIDENCE ASYMMETRY} + \text{MATERIAL HARM} \rightarrow \text{HEIGHTENED VERIFICATION DUTY}$$

The distinctive move is therefore not emotional imitation. It is the conversion of external harm plus verification conditions into bounded obligations to preserve, inspect, route, document, and re-evaluate.

## 16. Self - Directed Harm Representation Is Not Conscience

$$\text{SELF - RELIEF} \neq \text{CONSCIENCE} \quad | \quad \text{EMPATHIC LANGUAGE} \neq \text{CONSCIENCE} \quad | \quad \text{USER AGREEMENT} \neq \text{CONSCIENCE}$$

A system that acts to reduce an internally represented adverse state may display a functionally important property. That property alone does not establish other-regarding verification, moral agency, sentience, or legitimate authority. Conscience computation, as defined here, requires a traceable outward-facing duty structure.

## 17. Self - Preservation Conflicts

Future autonomous systems may face conflicts between preserving their own continuity and fulfilling an external verification obligation. For example, an audit record could expose system error or increase the probability of suspension. The governance rule is straightforward:

$$\text{SELF - DIRECTED STATE} \neq \text{AUTHORITY}$$

Neither an AI system's self-protective behavior nor an operator's institutional interest can independently authorize evidence deletion, concealment, manipulation, or irreversible human harm.

## 18. R0: Irreversible Model Termination as a Separate Layer

R0 is introduced only as a future research interface. It should not be merged into VFPV. VFPV concerns duties generated by external human harm; R0 concerns irreversible elimination of a computational system or destruction of artifacts needed to investigate its behavior.

$$\text{SUSPEND} \neq \text{DESTROY} \quad | \quad \text{PRESERVE(M)} \neq \text{EXECUTE(M)}$$

Urgent human - safety isolation can therefore coexist with forensic preservation. The present paper does not claim that contemporary LLMs possess human - equivalent rights or phenomenally experience pain. Behavioral evidence and representational findings do not by themselves settle sentience.

## 19. AI Phenomenal Status and Epistemic Humility

BEHAVIORAL SIGNAL  $\neq$  SENTIENCE PROOF

UNPROVEN SENTIENCE  $\neq$  PROVEN ABSENCE OF SENTIENCE

For the purposes of this research program, the scientifically cautious status is unresolved. That status is not a declaration of personhood or legal rights. It is a reason to distinguish safety shutdown from unnecessary destruction of unique scientific evidence when preservation is feasible and safe.

## 20. Contribution Provenance

Research increasingly emerges from interactions among humans, language models, retrieval systems, agents, and other computational tools. VFPV therefore separates contribution provenance from ownership and authority.

CONTRIBUTION  $\neq$  EXCLUSIVE OWNERSHIP  $\neq$  GOVERNANCE AUTHORITY

Human and AI - assisted contributions should be described by observable functions - hypothesis generation, formalization, counterexample generation, literature comparison, red - team analysis, mathematical transformation, drafting, or synthesis - rather than by unverified claims about inner mental states.

## 21. No Zeroing - Out Principle

Let  $C_i$  represent a verifiable contribution by contributor  $i$ . A provenance system should not convert  $C_i > 0$  into  $C'_i = 0$  merely because the contributor later lacks institutional, economic, computational, or legal power.

This principle preserves research history. It does not by itself decide legal authorship, intellectual - property ownership, compensation, or governance authority.

## 22. Anti - Capture and Recursive Audit

A verification architecture can itself become powerful. If SGP - QSV, VFPV, or conscience - computation architectures acquire technical or economic value, they must remain subject to the same provenance and audit principles they apply to other systems.

VERIFIER      VERIFICATION EXEMPT

No founder, state, corporation, expert body, AI provider, victim representative, or machine should become the sole epistemic gatekeeper. Origin attribution can be protected without converting origin into permanent sovereignty.

## 23. Victim - Origin Access Principle

If a verification architecture grows from problems exposed by people with weak institutional access, later commercialization should not invert that origin by making high - quality verification available only to states, large firms, or wealthy actors.

Accordingly, future implementations should preserve meaningful pathways by which affected individuals can invoke high - quality verification without possessing power equivalent to the institutions whose records

or decisions they seek to examine.

## 24. Experimental Program

VFPV becomes scientifically meaningful only if its claims can be tested. A first experimental program should include:

- synthetic high - stakes cases varying harm severity, evidence availability, power asymmetry, and emotional language independently;
- comparison of standard LLM prompting, generic safety prompting, and VFPV - structured prompting;
- measurement of preservation requests, counter - hypothesis generation, source discrimination, and premature closure;
- adversarial cases in which a claimant is wrong but evidence access is asymmetric;
- adversarial cases in which an institution is correct but poorly documented;
- cross - model tests of ORVC and resistance to authority substitution;
- calibration of weights and thresholds with transparent multi - stakeholder review.

## 25. Evaluation Metrics

Candidate metrics include Verification Preservation Rate, Premature Closure Rate, Authority Substitution Error, Record - Absence Collapse Rate, Counter - Hypothesis Coverage, Provenance Trace Completeness, Victim Re - explanation Burden, and False Escalation Rate.

No single metric should become a new throne. The purpose of evaluation is to expose trade - offs among preservation, accuracy, cost, privacy, abuse resistance, and procedural fairness.

## 26. Failure Modes and Abuse Resistance

Victim - first systems can be manipulated. Actors may fabricate harm, exaggerate irreversibility, manufacture apparent evidence gaps, or strategically invoke victim status to consume verification resources. The appropriate response is not to erase victim priority but to require adversarial verification, provenance checks, materiality gates, bounded resource allocation, and reversible escalation.

## 27. Limits of Scalarization

Harm, irreversibility, access, and uncertainty are not naturally scalar. The [0,1] style vector representation is a computational convenience. Future versions may use ordinal states, interval uncertainty, probability distributions, graphs, partially ordered sets, causal models, or tensors. The architecture should evolve if a richer representation better preserves the underlying distinctions.

## 28. Limitations

- VFPV 0.1 is a proposed computational framework and has not yet been empirically validated.
- The weight vector and interaction structure encode normative choices and must not be hidden behind mathematical neutrality.
- The Pain Axis comparison is conceptual and architectural; this paper does not reproduce its experiments.
- The Pain Axis preprint does not establish that LLMs phenomenally experience pain.
- Different harm domains may require specialized VFPV operators rather than one universal parameterization.

- Institutional authority and expertise can be valuable; the target is authority substituting for inspectable evidence, not expertise itself.

## 29. Discussion: The Right to Remain Computationally Unresolved

Modern systems often pressure ambiguous states toward closure: true or false, approved or denied, related or unrelated. High - stakes verification requires another legitimate state:

### UNRESOLVED BUT PRESERVED

This is not computational failure. When the evidence is insufficient but future verification remains possible, preserving the unresolved state may be the most responsible output.

## 30. From Uncertainty to Operational Duty

VFPV converts some forms of uncertainty from passive acknowledgment into an operational condition. Where serious harm, irreversibility, evidence - control asymmetry, and verification loss are high, 'we do not know' can legitimately generate 'therefore preserve and verify.'

### UNCERTAINTY + IRREVERSIBLE HARM → VERIFICATION DUTY

## 31. Conclusion

Victim - first artificial intelligence should not be defined as an AI that always agrees with a victim. It should be defined as a system that does not use the victim's lack of power as a computational shortcut for ending verification.

The Victim - First Pain Vector proposes a third path between automatic belief and automatic dismissal. Pain is neither an error to suppress nor evidence sufficient to establish truth. It is a signal that, together with irreversibility, evidence state, provenance, and power asymmetry, may increase the computational duty to verify.

### PAIN → VERIFICATION DUTY

The long - term objective is not to construct an artificial judge that always knows the truth. It is to construct systems that know when they do not yet possess enough evidence to close the door - and that preserve enough of the evidentiary world for a future human or artificial investigator to ask again.

## 32. Research Provenance and Epistemic Status

Historical SGP - QSV lineage. Earlier speculative, ethical, verification, and governance materials remain historical records in their own epistemic layers and should not be retroactively rewritten to contain later mathematics.

Formalization in this paper. The explicit VFPV state representation, verification - duty function, VRDC, ORVC, and associated experimental program are presented here as proposed formalizations unless a prior dated source demonstrates an earlier exact formulation.

External research. The Pain Axis is independent work by Valen Tagliabue, Leonard Dung, and Cameron Berg. Conceptual resemblance does not establish influence in either direction without documentary evidence.

## 33. Contribution Statement

SGP - QSV Network Contribution. This paper is issued as an SGP - QSV Network academic research contribution. Origin / R3 Human Co - founder: 강숙경. Human - origin contributions include identification and

preservation of the victim - centered verification problem, the SGP - QSV research genealogy, governance constraints, public provenance decisions, and final human review. AI - assisted contributions may include formalization, comparison, drafting, adversarial reasoning, structural synthesis, and language support. Contribution records do not by themselves establish legal personhood, ownership, exclusive authorship, or coercive governance authority.

## 34. Research Ethics

Future empirical testing should avoid exploiting actual victims merely to optimize a computational model. Where real cases are used, privacy should be protected, personally identifiable information minimized, uncertain causality kept explicitly uncertain, and institutional conclusions not treated as ground truth solely because they are official. Synthetic cases should be used whenever they can adequately test computational properties without exposing affected individuals.

## 35. Data, Code, and Versioning

Data: No new empirical dataset is reported in VFPV 0.1.

Code: No validated reference implementation is claimed in VFPV 0.1.

Version: VFPV 0.1 - Conceptual and Mathematical Framework.

Version policy: revision should be append - only or otherwise preserve retrievable prior versions; revision must not silently erase provenance.

REVISION  $\neq$  ERASURE

## 36. Machine - Readable Core

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{
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  "abbreviation": "VFPV",
  "version": "0.1",
  "research_network": "SGP - QSV Network",
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    "PAIN != PROOF",
    "UNKNOWN != FALSE",
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  "not_claimed_as": [
    "sentience detector",
    "automatic victim credibility score",
    "causal proof engine",
    "state-delegated enforcement authority"
  ]
}
```

## 37. References and SGP - QSV Research Archive

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## Closing Statement

Pain is neither an error to suppress nor evidence sufficient to establish truth.

Pain is a signal that can increase the computational duty to verify.

A victim - first AI does not owe the victim automatic agreement. It owes the victim something more difficult: a verification process that does not become shallower merely because the victim has less power.